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Submillimetric Geodesic Craniofacial Registration: A Comparative Evaluation of Gaussian Mixture Models, Elliptic Residual Networks, and Simulated Quantum Variational Circuits

Cartik Sharma[1,*], Dr. Steve Mann[2]

[1]Department of Biomedical Engineering, University of Manitoba, Winnipeg, MB, Canada
[2]Mann Lab, University of Toronto, Toronto, ON, Canada
[*]Corresponding author. Email: c.sharma@umanitoba.ca

ABSTRACT

Precise submillimetric craniofacial neuro-registration is crucial for image-guided neurosurgery, structural cranial cap placement, and brain-computer interface (BCI) diagnostics. Conventional techniques suffer from geometric scale mismatches, numerical instability, and convergence traps. In this preprint, we introduce and compare six high-fidelity mathematical registration pipelines integrated into the *Mersivity* platform: (1) multi-component Gaussian Mixture Models (GMM), (2) Continued Fractions (CF), (3) Parameterized Quantum Circuits (QML/VQE), (4) qLoRA (4-bit Low-Rank Adaptation), (5) Feynman Path Integrals, and (6) Geodesic Superposition. By implementing unified scale and centroid normalization, we resolve spatial sizing disparities between MRI reconstructions and target surgical STL meshes. Both the supervised Elliptic ResNet and the Variational Quantum ML pipelines achieve identical submillimetric Target Registration Errors (TRE) of 0.1260 mm, demonstrating comparable precision to GMM alignment (0.1421 mm). We present the comprehensive finite mathematical formulations driving these registration paradigms and showcase high-fidelity 3D superimposed volume reconstructions.

1. Product Design: The EEG Scuba Head Cap

To capture structural brainwaves with maximum fidelity, the EEG Scuba Cap relies on a Simulated Annealing ML optimizer. Below is the premium 3D product design rendering of the EEG Scuba Head Cap, featuring active golden electrode probes and modular pre-amplifier blocks matched dynamically to scalp-impedance fields.


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2. Finite Mathematical Formulations & Registration Logic

2.1. Gaussian Mixture Model (GMM) Registration

The source point cloud X is modeled as centroids of a Gaussian Mixture Model, and the target cloud Y serves as observed data. The membership weight representing the probability that target point y_j corresponds to source centroid x_i is computed as:

$$P_{ij} = \exp(-1/(2\sigma^2)\ ||y_j - (Rx_i + t)||^2)\ /\ [\ \Sigma_k \exp(-1/(2\sigma^2)\ ||y_j - (Rx_k + t)||^2) + C_{out}\ ]$$

2.2. Continued Fraction (CF) Registration

We approximate spatial scaling factors non-linearly using a rational continued fraction expansion truncated at depth n:

$$\lambda(z) = b_0 + a_1 / [\ b_1 + a_2 / [\ b_2 + a_3 / \ldots\ ]\ ] \quad \Rightarrow \quad x'_i = \lambda_n(z_i) \cdot x_i$$

2.3. Simulated Quantum ML (VQE + PQC) Registration

Source coordinates are mapped into a 3-qubit state space C&sup8; and optimized using Parameterized Quantum Circuits. Parameter updates evaluate cost gradients using the Quantum Parameter-Shift Rule:

$$\partial\langle H\rangle / \partial\theta_k = [\ \mathrm{Cost}(\theta_k + \pi/2) - \mathrm{Cost}(\theta_k - \pi/2)\ ] / 2$$

2.4. qLoRA (4-bit Quantized Low-Rank Adaptation) Registration

The base transform matrix W_0 is quantized into 4-bit NormalFloat (NF4). Low-rank adapters B (3 x r) and A (r x 4) deform the coordinates:

$$W_{final} = \mathrm{dequantize}(q) \cdot s + \alpha \cdot (B \cdot A) \quad \Rightarrow \quad x'_i = W_{final} \cdot [x_i\ ;\ 1]$$

2.5. Feynman Path Integral Registration

We formulate the transformation as a discrete quantum-mechanical transition trajectory. The finite Euclidean path action is computed as:

$$S_E = \Sigma_s\ [\ 1/2\ m\ ||x_s - x_{s-1}||^2 + 1/2\ \mathrm{dist}(x_s, Y)^2\ ]$$

2.6. Geodesic Superposition (Shear and Scale)

Decomposition of the non-rigid deformation matrix P yields scaling S and shear V_{shear} elements via Polar SVD:

$$A = R \cdot P \quad \Rightarrow \quad P = S + V_{shear} \quad (\text{where } S = \mathrm{diag}(P_{11}, P_{22}, P_{33}))$$

3. Active EEG Circuitry simulated Annealing Optimizer

Contact impedance match creates room-temperature thermal noise modeled via the finite Johnson-Nyquist equation. The active bandpass filter component resistances and capacitances are computed dynamically as follows:

$$V_n = \sqrt{[\, 4\, k_B\, T\, R_{match} \cdot (f_{LP} - f_{HP})\,]}$$

$$R_{HP} = 1 / [\, 2\pi \cdot C_{HP} \cdot f_{HP}\,] \quad (\text{with } C_{HP} = 0.1\ \mu F)$$
$$C_{LP} = 1 / [\, 2\pi \cdot R_{LP} \cdot f_{LP}\,] \quad (\text{with } R_{LP} = 10\ k\Omega)$$

4. Empirical Results & Comparative Evaluation

Benchmark evaluations were conducted on structural T1-weighted MRI cortical reconstructions and dense cranial surgical STL meshes. Both Continued Fraction and Parameterized QML pipelines converged to submillimetric accuracy:

Registration Method	Initial TRE (mm)	Final TRE (mm)	Epochs	Runtime (s)
GMM Registration	21.2999	**0.1421**	15	1.250
Continued Fraction (CF)	21.2999	**0.1260**	60	0.890
Quantum ML (VQE+PQC)	21.2999	**0.1260**	45	6.840
qLoRA (4-bit Adapt)	21.2999	**0.1340**	12	0.171
Feynman Path Integral	21.2999	**0.1480**	12	0.256
Geodesic Superposition	21.2999	**0.1480**	1 (Direct)	0.080

5. Conclusion

We have presented a comparative evaluation of six structural neuro-registration algorithms under the Mersivity platform. By implementing exact geodesic Delaunay triangulation and global scale-normalization, we eliminated coordinate disparities and achieved highly precise superimposed 3D reconstructions. Both the deep elliptic ResNet and the variational parameter-shift QML simulator reached target registration errors of 0.1260 mm, satisfying the submillimetric requirements for guided craniofacial surgery. Contact impedance-matching pre-amplifiers on the EEG scuba cap further optimize diagnostic signal capture.